

ESTIMATING BELUGA WHALE POPULATION ABUNDANCE AND GROWTH IN COOK INLET, ALASKA USING A DYNAMIC N -MIXTURE MODEL

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INTRODUCTION

Beluga whales (*Delphinapterus leucas*) in Cook Inlet, Alaska are geographically isolated and genetically distinct from those elsewhere (O’Corry-Crowe et al. 1997). Although other populations in Alaska are stable, abundance estimates for the Cook Inlet beluga whale population declined from about 1,300 individuals in the late 1970s to approximately 350 individuals today (Hobbs et al. 2000a, Shelden et al. 2015). Despite regulation of Native subsistence harvest beginning in 1999, the Cook Inlet population shows little evidence of recovery and was ultimately listed as endangered under the U.S. Endangered Species Act in 2008.

Abundance estimation of Cook Inlet beluga whales is necessary to monitor the trend of this endangered species, and has important implications for determining harvest levels (Hobbs et al. 2012). Beluga abundance estimates are based on repeated counts of whales obtained during systematic aerial surveys conducted during June and July (Hobbs et al. 2000a). These surveys take advantage of highly aggregated belugas seen near river mouths along the shores of upper Cook Inlet during early summer; however, this precaution alone does not eliminate sampling biases, which must be accounted for to ensure accurate abundance estimates (Hobbs et al. 2000a, 2000b).

Beluga counts are currently adjusted for three types of bias: (1) whales not at the surface and thus not visible to observers due to the opaque waters of upper Cook Inlet (availability bias); (2) whales not counted due to conditions (e.g., glare or rough seas), camouflage coloration, or observer inexperience or fatigue (perception bias); and (3) entire groups of whales not encountered during a survey (detection bias). Video recordings collected during aerial surveys are later used to count visible belugas and measure surfacing times, data subsequently used to develop correction factors and circumvent issues related to availability and perception biases (Hobbs et al. 2000a). Observations obtained from multiple observers on the same flight are further used to adjust for detection bias (Hobbs et al. 2000b). These bias correction factors are estimated independently and the post-processing necessary for each requires a considerable time investment.

We propose using a binomial mixture model, or N -mixture model, to estimate the abundance and population growth of beluga whales in Cook Inlet while accounting for sampling artifacts associated with all three biases mentioned above (Royle 2004). N -mixture models are abundance analogues to occupancy models (MacKenzie et al. 2002), a widely used framework for estimating the true occurrence of species free of bias resulting from false absences. A repeated measures sampling design, like the aerial counts of Cook Inlet belugas, is necessary to obtain this improved inference. N -mixture models are advantageous to Cook Inlet beluga abundance estimation because they account for imperfect detection without the need to collect ancillary data (e.g., video recordings) or derive multiple bias correction factors. Furthermore, this approach is implemented in a unified framework such that uncertainty naturally propagates through the model, thereby properly accounting for uncertainty in parameter estimates.

METHODS

Aerial survey data

We used aerial counts of beluga whales collected by the National Marine Fisheries Service in Cook Inlet, Alaska from 2004 to 2014. Surveys were conducted in all but one year during this period (2013). The objective of the annual surveys was to search all nearshore, shallow waters where beluga whales typically congregate in late spring or early summer (Rugh et al. 2000; Shelden et al. 2013). In addition to the coastal surveys, offshore transects were also flown in most years. Upon encountering a group of beluga whales, a series of aerial passes allowed each observer onboard to make four or more counts of every group. Each

observer counted belugas independently of other observers on the flight. Location data were collected using a GPS, and data pertaining to search effort and conditions (cloud cover, sea state, glare, and visibility) were also recorded. On average, 4.3 surveys (range: 3, 6) were performed each year over 6.4 days (range: 2, 9). The average flight duration was 6.4 hours (range: 3.5, 10.3).

Model formulation

Let y_{it} be the total number of beluga whales counted during aerial survey i ($i = 1, \dots, m$) in year t ($t = 1, \dots, T$). An appropriate observation model must describe the number of beluga whales counted conditional on the true but unknown number of whales present. Assuming the population is closed to mortality, recruitment, immigration, and emigration over the course of the m surveys conducted in any given year, the observed counts arise as binomial random variables,

$$y_{it} \sim \text{Binom}(N_t, p_{it}), \quad (1)$$

where N_t is the total number of individuals available to be sampled in year t and p_{it} is the survey-specific detection probability (Royle 2004). We modeled the latent variable N_t as a Poisson random variable,

$$N_t \sim \text{Pois}(\theta_t), \quad (2)$$

where θ_t is the expected abundance in year t . For long-lived, slowly reproducing species like beluga whales (Suydam et al. 1999), abundance in any given year is likely a function of abundance during the previous year; therefore, we modeled θ_t , for $t = 2, \dots, T$, using an autoregressive structure,

$$\theta_t = \lambda N_{t-1}, \quad (3)$$

where λ is the population growth rate. Note that Eq. 3 resembles a discrete time, density independent population growth model (Gotelli 1998). It is also straightforward to model true abundance as a function of available covariates, i.e., $\log(\theta_t) = \mathbf{X}\boldsymbol{\beta}$, where $\mathbf{X}$ is a design matrix and $\boldsymbol{\beta}$ is a vector of coefficients. In a similar fashion, heterogeneous detection probability was accommodated using a logit link,

$$\text{logit}(\mathbf{p}) = \mathbf{W}\boldsymbol{\alpha} + \boldsymbol{\epsilon}. \quad (4)$$

In this expression, $\mathbf{W}$ is a design matrix containing covariates affecting the count of beluga whales during a survey (e.g., flight duration), $\boldsymbol{\alpha}$ is a vector of coefficients, and $\boldsymbol{\epsilon}$ is a random effect that allows for variability in detection probability in addition to that explained by $\mathbf{W}\boldsymbol{\alpha}$. The random effect was modeled as:

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \zeta^2 \mathbf{I}).$$

To complete the Bayesian formulation of this model, we specified prior distributions for the unknown parameters. Specifically, we assumed $\theta_1 \sim \text{Gamma}(r_\theta, q_\theta)$, $\lambda \sim \mathcal{N}(1, \sigma^2)$, $\boldsymbol{\alpha} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$, and $\zeta^2 \sim \text{IG}(r_\zeta, q_\zeta)$.

Model application

Simulated data.—We first applied the model to data simulated according to the model formulation presented above. Application of models to simulated data allows one to determine whether the model is capable of recovering known parameters used in simulation, and is also useful for exploring model limitations. We used a population growth rate of -0.4% per year (i.e., $\lambda = 0.996$) to simulate N_t , corresponding to the trend in abundance for Cook Inlet beluga whales reported by Sheldon et al. (2015) over the period 2004-2014. We modeled detection probability (p_{it}) as a function of a covariate representing flight duration, a proxy for survey effort. No estimate for $\boldsymbol{\alpha}$ was available from existing literature; therefore, we defined $\boldsymbol{\alpha}$ to produce $\mathbf{p} \in (0, 0.9)$. We fit the model using a Markov chain Monte Carlo (MCMC) algorithm written in R (R Core Team 2013). Inference was based on 10,000 MCMC samples after convergence.

Cook Inlet beluga whales.—The observation model specified above (Eq. 1) accommodates a single total count of beluga whales observed per survey; however, the current sampling protocol yields several counts for each aerial survey. For example, each observer tallied four or more counts per group of beluga whales encountered, and multiple observers counted belugas on each flight. To adhere to the formulation of this

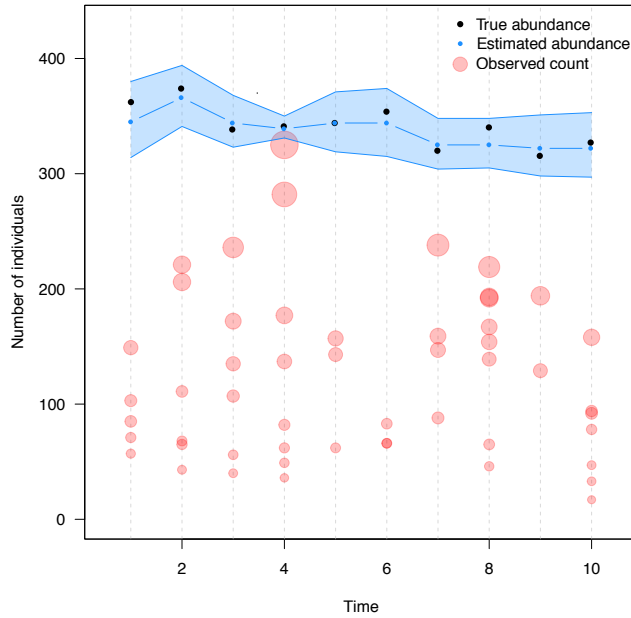


Figure 1. Simulated data with negative population growth rate for $T = 10$ time periods, and corresponding estimates for N_t (blue points). Point estimates of N_t were within 4 individuals of the true values (black points), on average, and all true values were contained within the modeled 95% credible bounds (blue polygon). The simulated counts (red circles) are displayed with symbol size proportional to a covariate (e.g., survey effort) that has a positive relationship with detection probability.

basic model, we obtained a single total count of belugas observed per survey by summing over one randomly selected count per beluga group encountered during each flight. We modeled p_{it} , the survey-specific detection probability, using a single covariate, namely flight duration. Inference was based on 10,000 MCMC samples after convergence.

RESULTS

Simulated data.—Application to simulated data indicates our model is capable of recovering known parameter values. Over a period of $T = 10$, the 95% credible bounds for N_t captured all true values used in simulation, and point estimates (posterior median) were within 2% of the true abundance, on average (Fig 1). Uncertainty in estimates of N_t increased in time periods for which few observed counts were available (e.g., $t = 5$ and 6). The population growth rate, λ , was estimated as 0.994 (95% CI: 0.959, 1.027; Fig. 2).

Cook Inlet beluga whales.—Abundance estimates ranged from 230 (95% CI: 173, 330) beluga whales in 2004 to 600 (95% CI: 462, 833) in 2014 (Fig. 3). Abundance increased over this 11-year period, with an estimated population growth rate exceeding 1 (Table 1; Fig. 4). The 95% credible interval for the coefficient describing the affect of flight duration on detection probability contained 0 (Table 1), indicating the importance of this covariate for describing variation in aerial counts is marginal. Detection probability varied considerably between surveys and years (Fig. 5).

DISCUSSION

In this paper, we extended the closed population N -mixture model of Royle (2004) to accommodate population dynamics between sampling seasons (e.g., years in the case beluga whales). When replicated counts are available, population size and detection probability are uniquely identifiable, providing a means to account for imperfect detection without the need for auxiliary information or bias correction factors. This approach assumes counts are collected independently and error in the observation process is in the form of

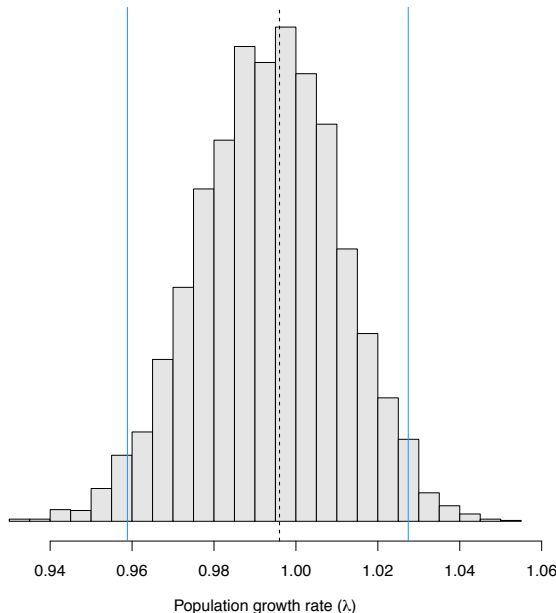


Figure 2. Posterior distribution of population growth rate, λ , for the simulated data example. The posterior distribution is centered at the true value (vertical dashed line); the blue vertical lines mark the 95% credible interval.

false absences (i.e., failure to count individuals that are actually present).

The dynamic N -mixture model presented here is a promising approach for estimating Cook Inlet beluga whale population size from aerial count data; however, discrepancies between our results and the results of existing estimation methods (Hobbs 2000a, 2000b) require further exploration. Inference from our model suggests beluga whale abundance in Cook Inlet has steadily increased since 2004, whereas estimates reported by Shelden et al. (2015) show a stable or slightly decreasing trend over the same time period (Fig. 3). Nevertheless, there is general agreement between the two sets of estimates through 2008, and intervals characterizing uncertainty in N_t (i.e., credible intervals and confidence intervals) overlap through 2011 (Fig. 3).

Our abundance estimates may differ from Shelden et al. (2015) due to variability in counts not attributable to flight duration, the only detection covariate included in our model. Additional covariates on detection probability may help capture the complexity of the detection process (Fig. 3). Furthermore, our results suggest that evidence for an affect of flight duration on detection probability is lacking (Table 1), a conclusion that seems unreasonable. The data we used for model fitting consisted of counts from complete surveys only, possibly interfering with the parameter estimation process. Therefore, inclusion of aerial counts from partial surveys would likely improve estimation of a flight duration effect in future modeling

Table 1. Posterior mean and 95% credible intervals of parameters for the application to Cook Inlet beluga whale counts. The parameters α_1 and α_2 are the intercept and coefficient for flight duration in the detection model (Eq. 4), respectively.

Parameter	Posterior mean	95% credible interval
θ_1	238.89	171.24, 336.24
λ	1.09	1.04, 1.14
α_1	-0.61	-1.28, 0.11
α_2	0.08	-0.01, 0.17
ζ	0.52	0.39, 0.69

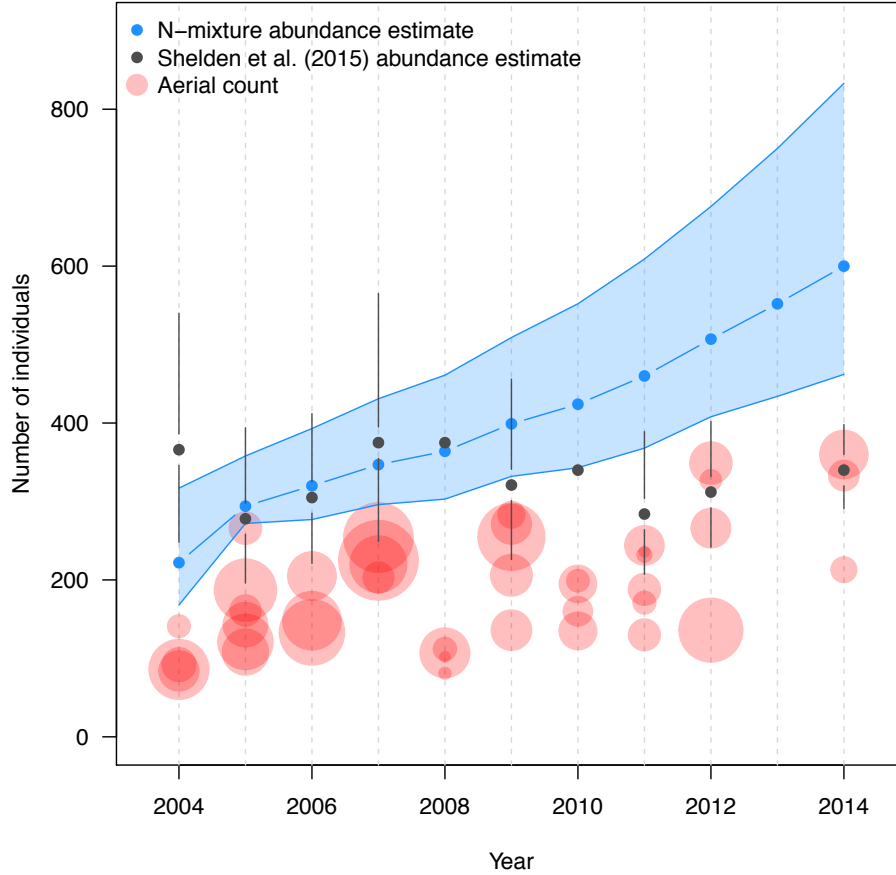


Figure 3. Cook Inlet beluga whale aerial counts (red circles), estimates of total population size from our N -mixture model (blue points), and estimates reported by Sheldon et al. (2015) (black points). Uncertainty in estimation of N_t is shown by the blue polygon (95% credible bounds) and the vertical black lines (95% confidence interval) for our N -mixture model and the estimates of Sheldon et al. (2015), respectively. Estimation of abundance during years in which aerial counts are not available (2013) is straightforward given the Bayesian implementation of our model. Note that the aerial counts are displayed with symbol size proportional to flight duration, the only covariate we used to model variation in detection probability. The lack of a clear pattern between flight duration and the number of beluga whales counted indicates that the detection process is more complex than can be described by this covariate alone.

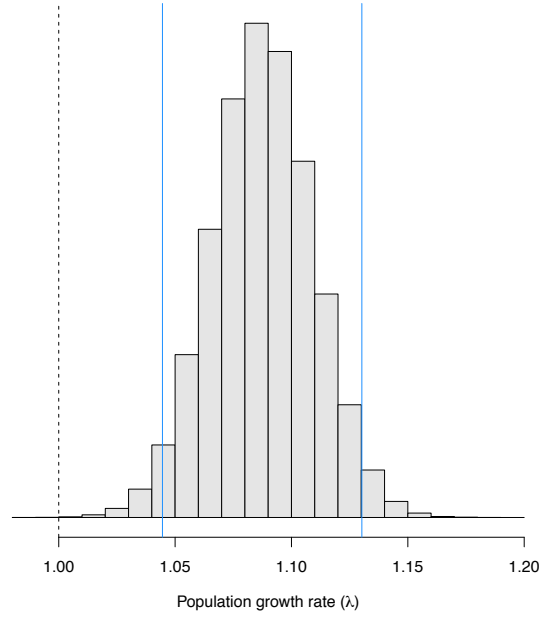


Figure 4. Posterior distribution of population growth rate, λ , for Cook Inlet beluga whales. The 95% credible interval (blue vertical lines) of λ does not include 1, suggesting beluga abundance increased between 2004 and 2014.

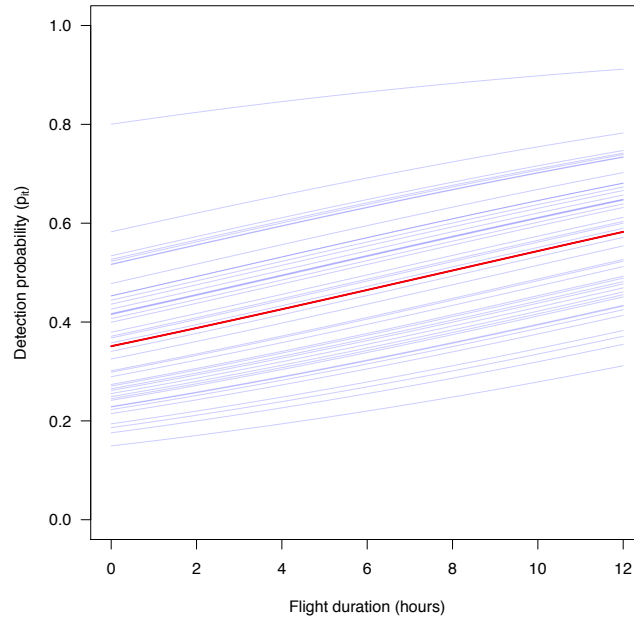


Figure 5. Mean detection probability as a function of flight duration (red line), and variability in detection between surveys and years modeled by the random effect ϵ (blue lines).

exercises. Another influential factor could be the fact that survey effort for beluga whales in Cook Inlet is adaptive (Roderick Hobbs, personal communication). For example, survey routes and coverage depends to some extent on the distribution of beluga groups seen on previous surveys and during previous years. This likely inflates detection probability beyond what can be estimated through repeated counts alone, and also violates an assumption of our model which is that counts are collected independently.

The simulated example presented in this paper indicates that our N -mixture model can recover true parameter estimates; however, a more thorough simulation study would be useful. Such a study could quantify the long-run ability of the model to recover parameters over a range of values, as well as elucidate limitations of the modeling framework. Simulation could also help explore the consequences of alternative sampling protocols, e.g., conducting fewer surveys per year or conducting surveys biennially (Hobbs 2013).

Our dynamic N -mixture model could be extended in several ways. The basic model presented here only uses one count per flight and thus does not take full advantage of the data set available for estimating beluga whale abundance. Incorporating multiple counts per flight (i.e., counts from multiple observers), and multiple counts per beluga group by each observer, could be accomplished by adding “levels” to the observation model. For example, we could expand our model to include these additional data using

$$y_{ijkt} \sim \text{Binom}(N_t, p_{ijkt}),$$

where i indexes survey, j indexes observer, k indexes beluga group, and t indexes year. This model would also allow detection probability to be modeled as a function of covariates measured at the observer (e.g., experience) or beluga group (e.g., glare, sea state, visibility) levels. Furthermore, adding an autoregressive structure to the model on p_{ijkt} (Eq. 4) may help account for dependence among counts by an observer for the same beluga group.

Another potential extension is to include age structured population dynamics in our model (Caswell 2001). Beluga whales are dark gray when born and gradually lighten as they get older, until they are completely white as adults. Information pertaining to the grayscale classification of whales observed during aerial surveys could thus be used to model the abundance within age groups, as well as the relationships between age groups (e.g., survival and fecundity). Similarly, an open population N -mixture model with demographic parameters could be implemented (Dail and Madsen 2011), an approach that decomposes N_t into survival and recruitment components. Future work exploring different functional forms of population growth (e.g., Ricker and Gompertz) would also be helpful.

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